

Jean-Louis Verdier

Des catégories dérivées des catégories abéliennes

Source-aligned English edition

Edited by Georges Maltsiniotis

Astérisque 239 (1996)

The thesis body was left unfinished by the author; no gap is silently completed.

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DERIVED CATEGORIES
OF ABELIAN CATEGORIES

Jean-Louis VERDIER

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A.M.S. Subjects classification

14F20, 18E30, 18E35, 18G05, 18G10, 18G15, 18G20, 18G35, 18G40, 55U15, 55U25, 55U30.

Jean-Louis VERDIER, 1977

Preface

Jean-Louis Verdier, who died tragically on 25 August 1989, defended his thesis for the *Doctorat d'État*, entitled *Derived Categories of Abelian Categories*, on 14 June 1967 in Paris. The present text is that of the typescript of this thesis, in its final version. The editors confined themselves to correcting minor errors. Only the part entitled *Introduction* had been deposited with the faculty. The body of the text was unfinished at the time of the defense, and remained so.

The genesis of this work is fairly well known (*cf.* [I]). We know that Grothendieck conceived the idea of derived categories at the beginning of the 1960s in order to provide the framework of homological algebra required for the vast generalization of the duality theorems that he had envisaged in the wake of his address to the 1958 International Congress in Edinburgh. He had proposed to Verdier, as a thesis subject, the construction of the formalism he had envisaged. In 1963, Verdier brought out a fascicle of results summarizing the essential content of the theory, *Catégories dérivées, quelques résultats (État 0)*, mimeographed by the IHÉS (reproduced in [SGA 4 1/2]). If the notion of a derived category is due to Grothendieck, it was Verdier who had earlier introduced that of a triangulated category. The axiomatic framework presented there, inspired by stable homotopy theory (*cf.* [P]), was to prove surprisingly fruitful.

It was only several years later that Verdier undertook a comprehensive account of his work. Why did he not finish it? Many hypotheses have been put forward. I would incline toward the following one. The part that is missing is the theory of derived functors. When Verdier reached that point in his writing, derived categories had already been used extensively in algebraic geometry. Various variants and generalizations were developing: the notion of pointwise derivability à la Deligne, nonadditive derived functors (homotopical algebra à la Quillen), filtered derived functors. From then on, the framework set by the results fascicle proved inadequate. Finding the right point of view would have required new foundational work which Verdier, interested at that time in other problems, had, it seems to me, neither the leisure nor doubtless the desire to carry out. As time passed, articles on derived categories appeared and their use spread, the formalism invading other fields of mathematics (such as linear partial differential equations). Completing his work then became, in Verdier's eyes, an increasingly unnecessary task ...

Incomplete though it is, Verdier's text nevertheless constitutes a valuable, indeed irreplaceable, reference (for example, the original and elegant presentation of spectral sequences given in it is, to my knowledge, found nowhere else). One can only rejoice that it has finally seen the light of day.

Paris, 27 April 1996

Luc Illusie

References

- [I] L. Illusie, *Catégories dérivées et dualité, travaux de J.-L. Verdier*, L'Enseignement Mathématique, 36 (1990), 369-391. (Talk delivered on 19 October 1989, during the ceremony in tribute to Jean-Louis Verdier organized by Université Paris VII.)

- [P] D. Puppe, *On the formal structure of stable homotopy theory*, Colloquium on Algebraic Topology, Aarhus Universitet (1962), 65-71.
- [SGA 4 1/2] “Cohomologie étale”, Séminaire de Géométrie Algébrique du Bois-Marie, SGA 4 1/2, by P. Deligne, Lecture Notes in Mathematics 569, Springer-Verlag (1977), 262-311.

Editors' Note

It was at the ceremony in tribute to J.-L. Verdier, organized by Université Paris VII on 19 October 1989, that M. Artin, P. Cartier, and L. Illusie conceived the idea of publishing J.-L. Verdier's thesis. Two years later I joined them and, with the support of J. Le Potier and M. Broué, obtained funding from Paris VII (URA 212) and the ÉNS (DMI) to key this text into $\text{\TeX}$. Once the typing was completed, the result was carefully compared with Verdier's typewritten original. A mathematical review still remained to be carried out, since Verdier had never made the final corrections.

This careful reading was carried out, with the utmost care, by P. Cartier (Chapter I), B. Keller (Chapter III), and, for the text as a whole, by M. Zisman and myself. Apart from the many typographical errors that do not merit enumeration, the principal modifications made are as follows. In Chapter I, Proposition (3.1.5) asserted an equivalence. This equivalence was replaced by an implication, since the converse (never used subsequently) is false. Proposition (3.4.1) contained a sign error, giving rise to sign errors in Chapter III ((1.3.4), (b), and (3.2.7.3)). This led to an adjustment of the definition of the “opposite triangulated category” (Chapter II, 1.1.7), so that the following assertion would remain true: *the derived category of the opposite category is the triangulated category opposite to the derived category*.

In Chapter II, the order of the propositions in §1.2 was changed. Indeed, after writing the text Verdier had realized that the assertion that the direct sum of two distinguished triangles is a distinguished triangle is a consequence of the other axioms for triangulated categories. He had therefore removed it from the list of axioms without making the resulting rearrangements. In addition, paragraphs (2.3.6) and (2.3.7), which were not used later and whose verification failed, were deleted. The definition of the notion of a stationary spectral object (4.4.2) was slightly modified, the original definition being too weak to establish the ensuing Theorem (4.4.3). It was verified that this had no effect on what followed.

The index of notation and the terminological index were completed. By contrast, the bibliography was left unchanged (except that the definitive references in the “Lecture Notes” were given for the SGA volumes). In particular, the absence of reference number 13 was not remedied.

I would like to express my warm thanks to M. Broué, A. Bruguières, P. Cartier, L. Gruson, L. Illusie, B. Keller, J. Le Potier, and M. Zisman, as well as to Université Paris VII, École Normale Supérieure, and Institut des Hautes Études Scientifiques, for the various ways in which they helped bring this project to fruition. I also thank C. Defosse and C. Gourgues for patiently keying this work into AMS- $\text{\TeX}$.

I would have liked to be able to extend these thanks to L. Doustaing, who died prematurely in December 1994.

Paris, 14 June 1996
Georges Maltsiniotis